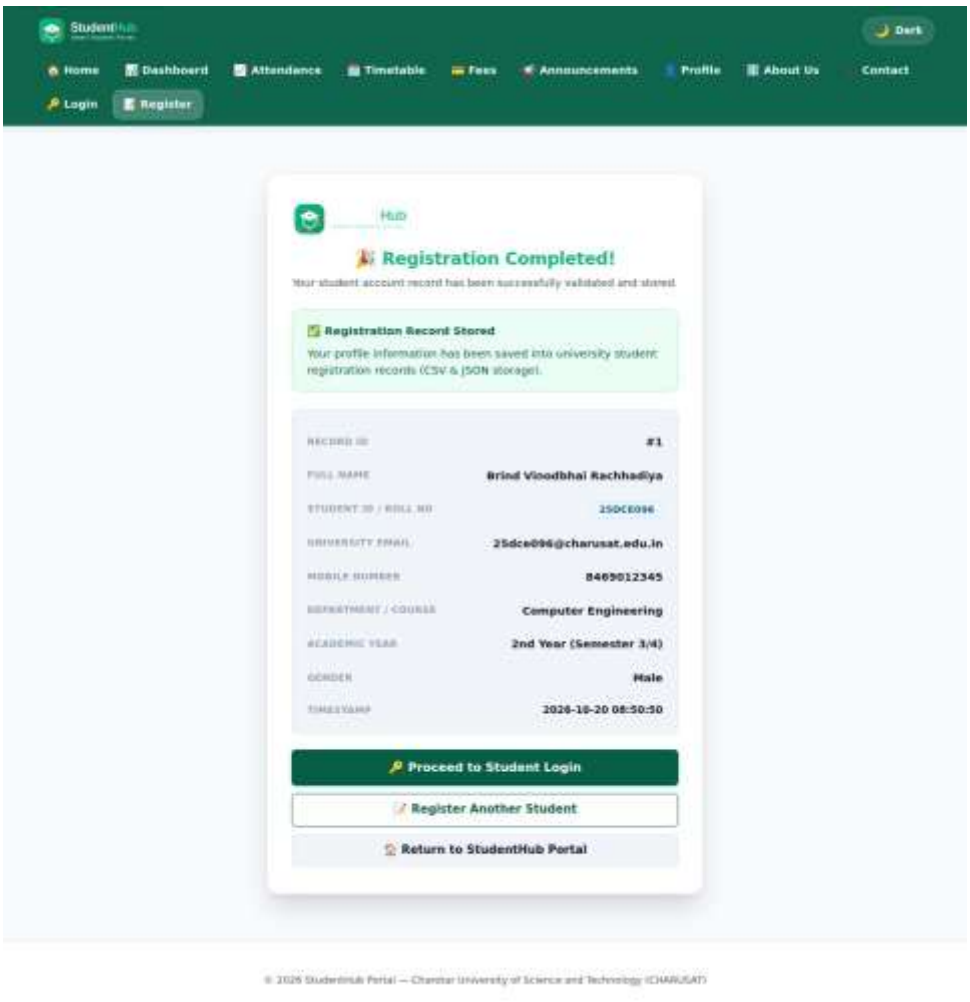
PRACTICAL-7

AIM: Process the student registration and contact forms on the server with PHP. Read the POST data, validate and sanitize every field, store valid records in CSV and JSON files, and show clear success or error messages.

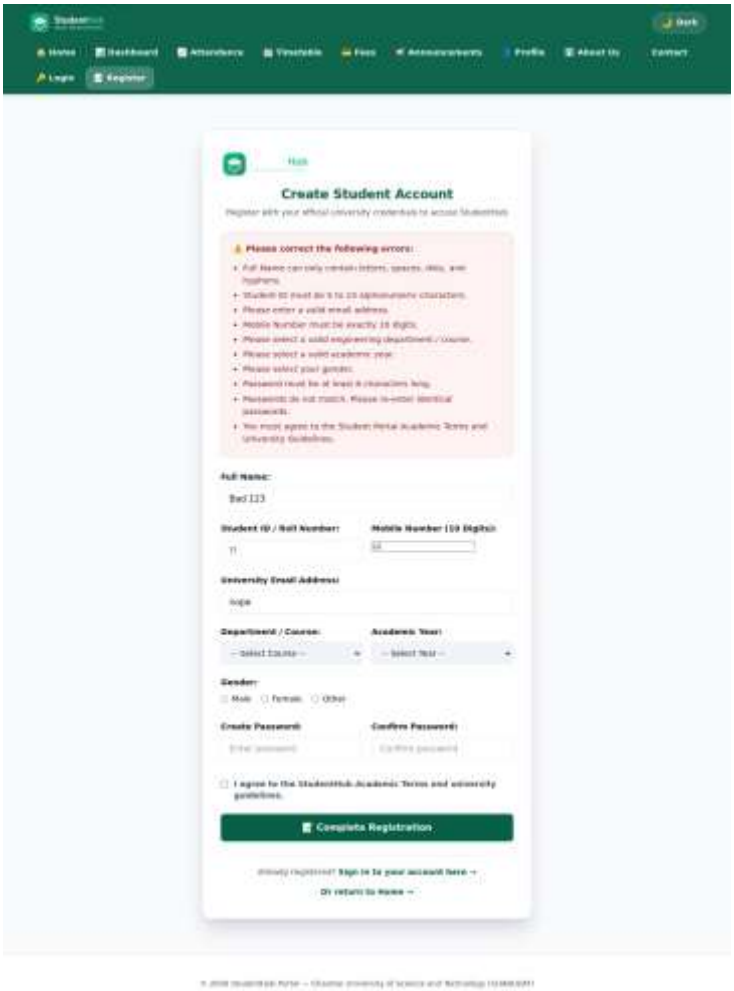
SOURCE CODE :

https://github.com/rachhadiyabrind-del/WDF_B/tree/main/Practical(1-4)(Studenthub)/Practical-7

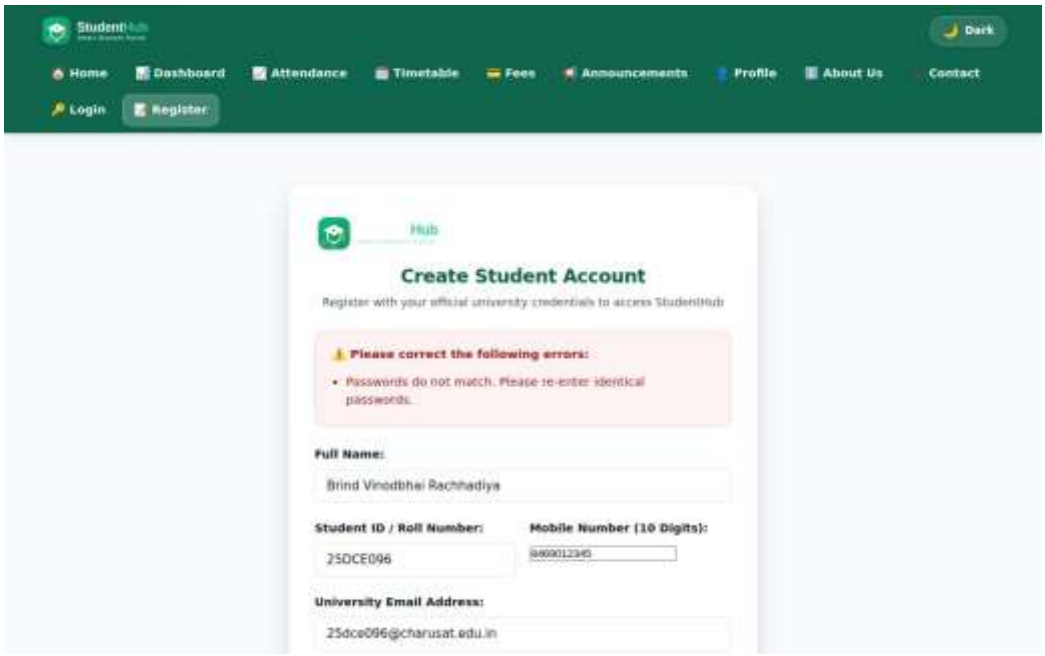
PREVIEW:



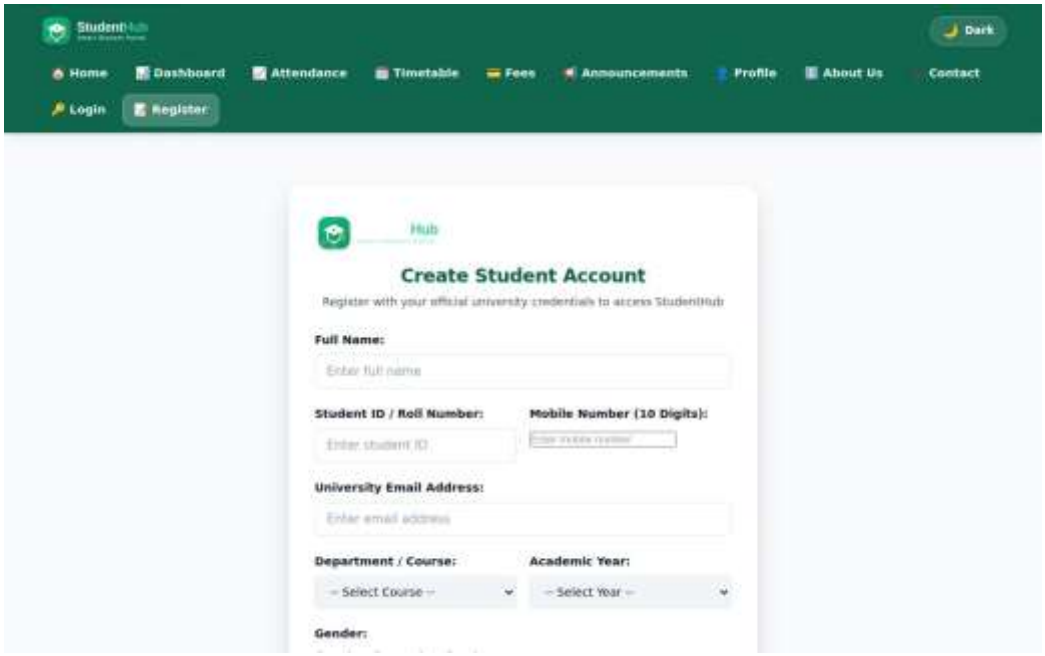
Valid registration - success page with the stored details



Invalid submission - server-side error list



Password and confirm password do not match



GET request - empty form, nothing is stored

Practical-7/registrations.csv

```
id,fullname,student_id,email,mobile,course,year,gender,registration_date
1,"Brind Vinodbhai Rachhadiya",25DCE096,25dce096@charusat.edu.in,8469012345,"Computer Engineering","2nd Year (Semester
3/4)",Male,"2026-10-20 08:50:50"
2,"Riya Shah",25DIT034,riya.shah@charusat.edu.in,9825123456,"Information Technology","2nd Year (Semester 3/4)",Female,"2026-10-
20 08:50:51"
3,"Karan Mehta",25DAI011,karan.mehta@charusat.edu.in,9724567890,"AI & Data Science","2nd Year (Semester 3/4)",Male,"2026-10-20
08:50:53"
```

registrations.csv after three submissions

Practical-7/registrations.json


```
[
    {
        "id": 1,
        "fullname": "Brind Vinodbhai Rachhadiya",
        "student_id": "25DCE096",
        "email": "25dce096@charusat.edu.in",
        "mobile": "8469012345",
        "course": "Computer Engineering",
        "year": "2nd Year (Semester 3/4)",
        "gender": "Male",
        "registration_date": "2026-10-20 08:50:50"
    },
    {
        "id": 2,
        "fullname": "Riya Shah",
        "student_id": "25DIT034",
        "email": "riya.shah@charusat.edu.in",
        "mobile": "9825123456",
        "course": "Information Technology",
        "year": "2nd Year (Semester 3/4)",
        "gender": "Female",
        "registration_date": "2026-10-20 08:50:51"
    },
    {
        "id": 3,
        "fullname": "Karan Mehta",
        "student_id": "25DAI011",
        "email": "karan.mehta@charusat.edu.in",
        "mobile": "9724567890",
        "course": "AI & Data Science",
        "year": "2nd Year (Semester 3/4)",
        "gender": "Male",
        "registration_date": "2026-10-20 08:50:53"
    }
]
```


registrations.json after three submissions

TEST CASES :

#	Input / action	Expected result	Result
1	Submit a valid registration form	Success message; new record appended to registrations.json and registrations.csv with the next ID	Pass
2	Submit the registration form with blank or invalid fields	All validation errors listed together; nothing stored	Pass
3	Enter a password and confirm password that differ	"Passwords do not match" error; no record saved	Pass
4	Open process_signup.php directly (GET request)	Empty form shown; no POST processing or file writing	Pass
5	Submit the contact form with valid details	Ticket number shown; row added to contacts.csv and contacts.json	Pass
6	Submit contact form with invalid e-mail and a message under 10 characters	Specific error for each invalid field; nothing stored	Pass
7	Inspect registrations.csv after several submissions	Header row written once; one row per record	Pass
8	Inspect registrations.json after several submissions	Valid JSON array; no password stored in the file	Pass

KEY QUESTIONS :

1. Is the form submitted using POST?

Yes. Both processors check the request method before reading any data. A normal page visit (GET) only shows the empty form, so nothing is stored by accident.

```
$isPost = ($_SERVER['REQUEST_METHOD'] === 'POST');
```

2. Are inputs validated and sanitized on the server side?

Yes. Sanitizing happens first: trim() removes spaces, strip_tags() removes HTML from text fields, FILTER_SANITIZE_EMAIL cleans the e-mail, and preg_replace() keeps only digits in the mobile number. Validation then checks each field: required fields, length limits, a regular expression for the name and roll number, FILTER_VALIDATE_EMAIL for the e-mail, and in_array() whitelists for course, year and gender so only known values are accepted.

```
$email = trim(filter_var($_POST['email'] ?? '', FILTER_SANITIZE_EMAIL));
if (!preg_match('/^[0-9]{10}$/', $mobile)) { $errors[] = '...'; }
```

The password is checked for length and matching, but it is never written to the CSV or JSON file.

3. Is file writing handled safely?

Yes. The file paths are built with __DIR__ and fixed file names, so user input never decides where a file is written. The JSON file is written with LOCK_EX to prevent two requests from writing at the same time. The CSV file is opened in append mode and locked with flock() while the row is written; the header row is added only when the file is new or empty. fputcsv() escapes commas and quotes inside the values. If any write fails, the user sees an error instead of the data being lost silently.

4. Are success and error responses displayed clearly?

Yes. All validation problems are collected into an $errors array and shown together in a red alert box with a bulleted list, so the user can fix everything in one pass. A successful submission shows a green confirmation with the ticket number and the stored values. Every value is printed through htmlspecialchars(), so entered text cannot inject HTML or script into the page.